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Revealing and overcoming fairness confusion in out-of-distribution detection

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ABSTRACT

Out-of-Distribution (OOD) detection prevents models from misclassifying OOD data that fall outside the in-distribution (ID) classes as ID categories. However, existing OOD detection methods ignore a critical metric, i.e., fairness metric. This oversight could result in unreliable predictions due to sensitive attributes in the data. To fill this gap, we introduce a novel and challenging problem termed *Fair OOD Detection* in this paper, which simultaneously considers OOD detection and bias induced by *Fairness Confusion* (FC) caused by sensitive attributes and their induced *Feature Shifts* (FS). Furthermore, we propose a novel metric termed Fair-OOD to identify FC phenomena in OOD detection, and a theoretically guaranteed semi-supervised solution named *Predictive Adaptive Calibration* (PACT) to simultaneously enhance OOD detection capability, ensure fairness, and mitigate FC without requiring the label of sensitive attribute for OOD data. Extensive experiments demonstrate that: (a) Fair-OOD can identify FC issues in models that existing fairness metrics fail to detect; (b) PACT effectively improves OOD detection performance while eliminating both FC and unfairness issues.

1. Introduction

Machine learning (ML) classification models trained under the closed-world assumption inevitably encounter out-of-distribution (OOD) inputs from unknown categories in real-world applications. As a result, OOD detection serves as a critical safety mechanism by identifying OOD samples while maintaining accurate classification on in-distribution (ID) data, thereby ensuring reliable model performance beyond the training distribution (Wang et al., 2025). Existing research in OOD detection primarily focuses on two directions: developing more efficient fine-tuning strategies (Hendrycks et al., 2019; Liu et al., 2020; Wang et al., 2023a), and designing effective scoring functions that better distinguish ID from OOD data (Hendrycks & Gimpel, 2017; Wang et al., 2023b).

Despite the notable progress in building robust and trustworthy ML systems, we observe that concurrent OOD detection approaches overlook another relative but important constraint, i.e., the fairness metric (Hardt et al., 2016; Jung et al., 2024), which requires equal predictions across different groups decided by the *sensitive attribute*. As illustrated in Fig. 1, the major and minor sample groups governed by the sensitive attribute (e.g., image background) exhibit similar distribution division patterns to those in OOD detection tasks, i.e., ID and OOD distributions (Wang et al., 2025). Unfortunately, as recent researchers have been aware of the frequent co-occurrence of the sensitive attribute within realistic data (Nam et al., 2020), this *relative but divergent* variable tends to confuse distinguishing between ID and OOD data, thereby undermining the performance of OOD detection. To be specific, we characterize this unexpected but commonly existed phenomenon as *Fairness Confusion* (FC), featuring the negative impact of the sensitive attribute from two folds. First, as illustrated in Fig. 1, the sensitive attributes themselves might yield direct, biased impact on OOD detection (e.g., the OOD horse is misclassified as a tiger or sika deer due to background). Moreover, sensitive attributes may further induce *Feature Shifts* (FS) and yield indirect impact by yielding feature disparities similar to the disparities between ID and OOD distributions. Specifically, these two effects are connected through the sensitive attribute: it can either directly affect the OOD prediction by serving as a shortcut cue, or indirectly affect the prediction by first inducing feature shifts and then disturbing the learned representation.

Different from traditional OOD distribution shifts, which usually focus on covariate shifts such as changes in image style or environment, or semantic shifts such as unseen object categories, FC arises when sensitive-attribute-related variations are entangled with distributional changes relevant to OOD detection. In such cases, the model may mistakenly regard sensitive cues or their induced feature shifts as valid evidence for OOD detection. Leaving this issue unresolved undermines the reliability of OOD detection, making its resolution essential.

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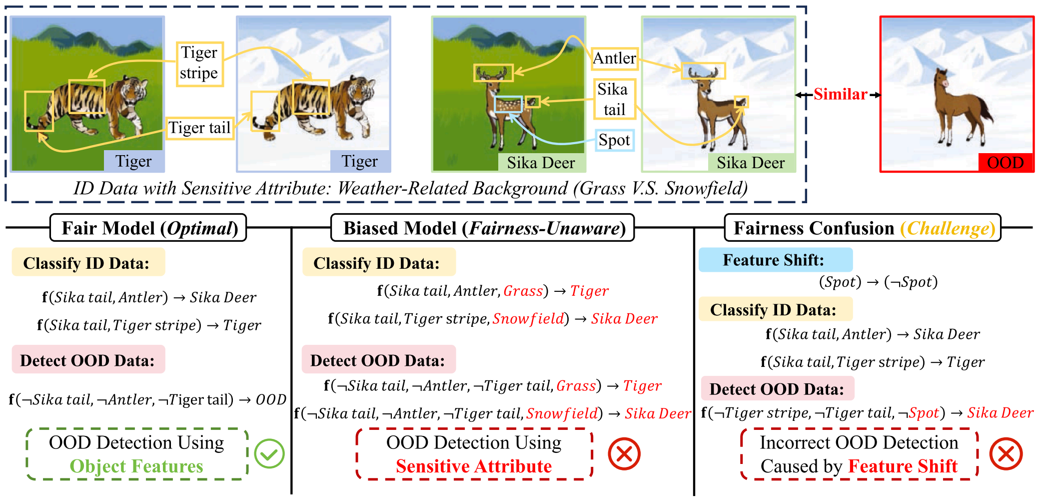


Fig. 1. Comparison of Fair Models (optimal), Biased Models (fairness-unaware), and Fairness Confusion (challenge) in OOD Detection. The symbol ¬ denotes logical NOT and **f** represents the model. Red font indicates erroneous evidence that should not be relied upon, or incorrect results. In OOD detection, besides the unfairness caused by sensitive attributes, a distinct fairness confusion issue emerges from feature shifts induced by sensitive attributes. However, current OOD detection literature overlooks both fairness and fairness confusion, a gap that motivates our work.

To migrate the fairness confusion, we thereby introduce a novel and challenging problem termed *Fair OOD Detection* (cf. Problem 1) in this paper, which simultaneously considers both the sensitive attributes of data and the *Feature Shifts* (FS) induced by these attributes. To the best of our knowledge, our work presents the first study on constructing fairness-aware OOD detection framework. Furthermore, as our analysis reveals that the fairness metrics fails to capture the impact of FS, their direct adaptation is limited. As shown in our example in Fig. 1, even when a model satisfies existing fairness metrics with respect to sensitive attributes themselves (e.g., backgrounds), it may still produce erroneous detection outcomes due to FS. To handle this challenge, we subsequently introduce the Fair-OOD metric (cf. Definition 1) to identify the FC phenomenon in OOD detection. It works by evaluating the prediction consistency of a model under varying sensitive attributes and FS conditions.

To address our proposed fair OOD detection problem, we subsequently propose a novel semi-supervised approach termed *Predictive Adaptive Calibration* (PACT), which is designed to achieve three key objectives: (a) enhancing the OOD detection capability of model, (b) ensuring fairness, and (c) mitigating FC. Specifically, PACT employs *Feature Distribution Regularization* (FDR) to align feature distributions within the same class while increasing the divergence of distributions across different classes. Concurrently, it utilizes *Predictive Distribution Calibration* (PDC) to amplify the discrepancy between the prediction distributions of ID and OOD data when the label of sensitive attribute in OOD data are unavailable. Furthermore, we conduct theoretical analyses and provide formal guarantees for both FDR and PDC. We summarize our main contributions as follows:

- We first incorporate the consideration of fairness issues in OOD detection and introduce a novel problem, Fair OOD Detection, by considering the issue, i.e., *Fairness Confusion* (FC).
- We introduce a novel metric, termed Fair-OOD, which is designed to identify and characterize the FC issue in OOD detection in a principled manner.
- We propose a novel algorithm, *Predictive Adaptive Calibration* (PACT) (cf. Algorithm 1), which simultaneously ensures model fairness, enhances OOD detection capabilities, and mitigates FC issue. And we provide theoretical guarantees for it.
- Extensive experiments on real-world datasets demonstrate that: (a) Our proposed metric, Fair-OOD, effectively identifies FC problems; and (b) Our proposed method, PACT, improves OOD detection performance while eliminating both FC and unfairness issues.

2. Empirical study: fairness confusion

In this section, we conduct a case study to investigate the following questions:

- How does sensitive attribute affect OOD detection?
- How does feature shift (FS) affect OOD detection?

2.1. Notations

Let $\mathcal{X}$ denote the feature space and $\mathcal{Y} = \{1, \cdots, c\}$ denote the ID label space. To facilitate formal analysis, we introduce a binary variable $Z \in \mathcal{Z} = \{i, o\}$ to indicate ID ($Z = i$) or OOD ($Z = o$) for $X \in \mathcal{X}$. Furthermore, as discussed in the introduction, the model may be influenced by sensitive attributes and their induced feature shifts. To formalize this, we denote the sensitive attribute and feature shift by the variables A and S, respectively. We assume A and S to be binary, i.e., $A \in \{a, a'\}$ and $S \in \{s, s'\}$, where s indicates the presence of FS and s' indicates its absence, and it may only occur when $A = a'$, as illustrated in Fig. 2, while our discussion framework can be easily generalized to categorical sensitive attributes and feature shifts. We consider the ID dataset

$$D_{\mathrm{I}} = \{\mathbf{x}_j^{\mathrm{I}}, A_j^{\mathrm{I}}, S_j^{\mathrm{I}}, Z_j^{\mathrm{I}}, Y_j^{\mathrm{I}}\}_{j=1}^{n} \overset{\text{i.i.d.}}{\sim} P_{\mathrm{I}}, \tag{1}$$

where $P_{\mathrm{I}} = P(X, A, S, Z, Y)$ is the ID joint distribution. And we consider a realistic scenario where both sensitive attributes and FS label variations in OOD data are unobserved, i.e., the OOD dataset is

$$D_{\mathrm{O}} = \{\mathbf{x}_k^{\mathrm{O}}, z_k^{\mathrm{O}}\}_{k=1}^{m} \overset{\text{i.i.d.}}{\sim} P_{\mathrm{O}}, \tag{2}$$

where $P_{\mathrm{O}} = P(X, Z)$ is the OOD joint distributions. Moreover, we consider the classification model consisting of two components: a feature extractor $\mathbf{f} : \mathcal{X} \to \mathbb{R}^d$, where $\mathbb{R}^d$ is the feature space, and a prediction function $\mathbf{h} : \mathbb{R}^d \to \mathbb{R}^c$ that outputs softmax probabilities.

2.2. A real-world case study

We conduct the case study by employing the BAR (Nam et al., 2020), a real-world action recognition dataset. Additional implementation details are provided in Appendix B.1.

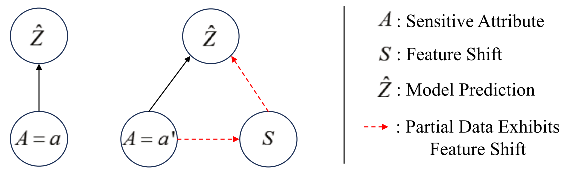


Fig. 2. Effect of sensitive attribute and feature shift (FS).

For the first question: Although a model achieves fairness in ID classification, sensitive attributes can still introduce bias in OOD detection, resulting in disproportionately higher error rates among marginalized groups. As shown in Fig. 3(a), the misclassification rate on ID data is identical for all sensitive-attribute values. Nevertheless, the model exhibits disparate OOD detection error rates across sensitive attribute values, revealing clear unfairness.

For the second question: FS-induced detection errors are predominantly observed among marginalized groups ($A = a'$). We measured the error rates of the model within this group under two FS conditions, i.e., with FS ($S = s$) and without FS ($S = s'$). As illustrated in Fig. 3(b), FS leads to significantly higher error rates in OOD sample detection compared to scenarios without such shift.

3. Fair OOD detection

This section formalizes the problem of fair OOD detection and shows, through counterexamples, that current fairness metrics cannot properly address fairness-confusion issues.

3.1. Problem setup

To address the challenging issue of incorporating fairness concerns related to sensitive attributes and FS into OOD detection, we need to solve the following critical problem.

Problem 1 (Fair Out-of-Distribution Detection)**.** Given a model trained on ID data D_I, the objective of fair OOD detection is to achieve two goals: (a) reliable predictive performance, such that samples from P_O are correctly identified as OOD and samples from P_I are classified into their correct ID classes; and (b) group-level fairness with respect to sensitive attributes and feature shifts.

Furthermore, we propose a novel metric, termed Fair-OOD, that jointly accounts for both sensitive attributes and FS to measure unfairness and FC within a unified framework.

Definition 1 (Fair-OOD)**.** In OOD detection, a fair model should simultaneously eliminate both discrimination arising from sensitive attributes A and the effects of the feature shifts S, i.e., the model should satisfy:

$$P(\hat{Z} = \hat{z}|A = a, S = s'_a, Z = z) = P(\hat{Z} = \hat{z}|A = a', S = s'_{a'}, Z = z),\quad \hat{z}, z \in \mathcal{Z}, \tag{3}$$

and

$$P(\hat{Z} = \hat{z}|A = a', S = s_{a'}, Z = z) = P(\hat{Z} = \hat{z}|A = a', S = s'_{a'}, Z = z),\quad \hat{z}, z \in \mathcal{Z}, \tag{4}$$

where $S = s_a$ and $S = s_{a'}$ represent the values of S when $A = a$ and $A = a'$, respectively. Here, z denotes the ground-truth ID/OOD label, whereas $\hat{z}$ denotes a possible model prediction. Although the above definition can be stated for any pair $\hat{z}, z$, in our Fair-OOD metric we mainly focus on the correct-decision case $\hat{z} = z$. This is because, under the same ground-truth label $Z = z$, only the case where the prediction satisfies $\hat{Z} = z$ indicates that the model makes the correct ID/OOD decision despite variations in sensitive attributes or the presence of feature shifts, thereby suggesting that the model does not suffer from unfairness or fairness confusion.

Remark 1. Fair-OOD characterizes group-level fairness by measuring differences in the conditional probabilities of correct ID/OOD decisions across sensitive groups and FS conditions. Eq. (3) quantifies the effect of sensitive attribute when no FS occurs ($S = s'$), while Eq. (4) measures the impact of FS in the data ($A = a'$) where FS may occur.

3.2. Comparison to other fairness notions

In the context of OOD detection, we demonstrate that prior fairness concepts fall short to measure the fairness confusion issue. To illustrate this, we compare with the fairness metric from fair-aware outlier detection because, in existing fairness research, outlier detection is the task most similar to OOD detection. We provide the formal definition of fair-aware outlier detection as follows.

Definition 2 (Fairness-aware Outlier Detection)**.** (Shekhar et al., 2021) A model for outlier detection is considered fair if it satisfies both the following fairness metrics:

- **Demographic Parity** (DP):
$$P(\hat{Z} = o|A = a) = P(\hat{Z} = o|A = a'); \tag{5}$$
- **Equal Opportunity** (EO):
$$P(\hat{Z} = o|A = a, Z = o) = P(\hat{Z} = o|A = a', Z = o). \tag{6}$$

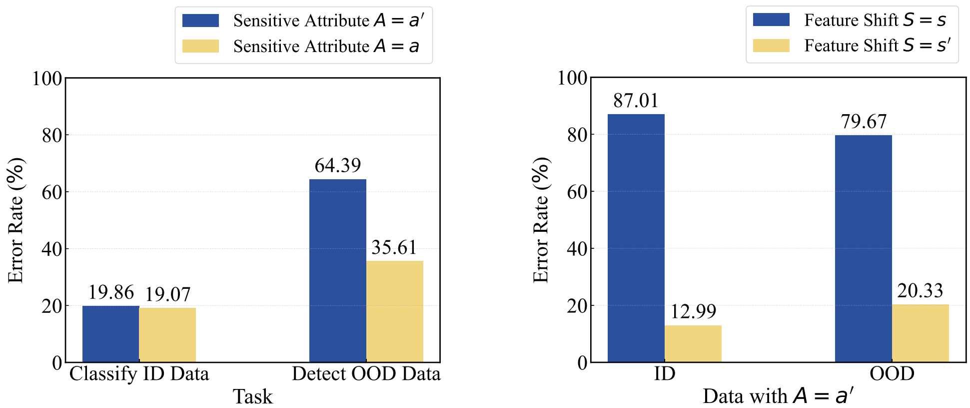
Remark 2. For consistency in presentation, we retain the variable Z throughout our analysis. It should be noted that in the outlier detection context, $Z = o$ indicates the label of an outlier, while $Z = i$ denotes the label of an inlier.

Through a counterexample, we demonstrate that conventional fairness metrics (e.g., DP and EO) fail to reliably detect unfairness in OOD detection due to their inability to address fairness confusion induced by feature shifts. Consider the scenario where $P(S = s) = P(S = s') = 0.5$, $P(S = s'|A = a) = 0.99$, and $P(S = s'|A = a') = 0.01$. We construct an unfair model $\mathbf{f}$ with $P(\hat{Z} = o|A = a, S = s) = 0.1$, $P(\hat{Z} = o|A = a', S = s') = 0.9$, $P(\hat{Z} = o|A = a =', S = s) = P(\hat{Z} = o|A = a, S = s') = 0.5$. Consequently, we obtain that $P(\hat{Z} = o|A = a) \approx 0.5 = 0.5 \approx P(\hat{Z} = o|A = a')$ (see Appendix A for the details). Therefore, DP cannot identify unfairness in this scenario. Similarly, EO, being a finer-grained variant of DP, also fails to detect unfairness. This is because these metrics solely consider the effects of the sensitive attribute, overlooking the impact of feature shifts. Distinct from them, our framework explicitly accounts for feature shifts to address the induced fairness confusion.

Moreover, conventional conditional fairness generally evaluates parity across sensitive groups after conditioning on selected variables, while subgroup fairness assesses disparities over predefined or intersectional subpopulations. Fair-OOD differs from these notions by organizing its comparisons according to the specific failure mechanism of fairness confusion in OOD detection. It first compares different sensitive groups under the common no-FS condition to measure the disparity associated with the sensitive attribute, and then compares samples with and without FS within the marginalized group to measure the additional disparity induced by FS. By integrating these two complementary comparisons, Fair-OOD distinguishes sensitive-attribute-related disparity from feature-shift-induced fairness confusion, enabling the identification of cases in which the disparity between sensitive groups is small but FS still substantially affects OOD decisions.

3.3. The need for dedicated treatment of fairness confusion

We further discuss why fairness confusion (FC) cannot be directly resolved by existing debiasing methods. Most fairness-aware learning methods are designed for ID classification (Edwards & Storkey, 2016; Madras et al., 2018), where the main goal is to reduce the dependence between model predictions and the sensitive attribute A (Zemel et al., 2013). However, fair OOD detection requires a different objective: the



(a) Effect of Sensitive Attribute (b) Effect of Feature Shift

Fig. 3. Experimental results of our case study.

model should not only reduce the influence of A, but also avoid confusing sensitive-attribute-induced feature shifts S with the distributional discrepancy between ID and OOD samples. Therefore, simply applying existing debiasing methods to OOD detection is insufficient.

Reweighting methods adjust the importance of samples from different sensitive groups (Chai & Wang, 2022; Kim et al., 2022). Although this can reduce group imbalance in ID classification, it does not explicitly constrain the feature space or the OOD decision boundary. In particular, reweighting cannot guarantee that samples from the same ID class but with different FS conditions are mapped to compact representations (Cheng et al., 2024; Roh et al., 2023). Moreover, in the fair OOD detection problem, the sensitive-attribute and FS labels of OOD data are unavailable, making it difficult to construct reliable group-wise weights for OOD samples.

Adversarial debiasing and fair representation learning also have limitations (Hwang et al., 2022; Jung et al., 2024; Lee et al., 2021). These methods typically suppress sensitive-attribute information or align representations across sensitive groups (Grari et al., 2024; Jones et al., 2025). However, FS is not necessarily identical to the sensitive attribute itself. It can appear as a feature-level variation induced by A and may still affect OOD scores even when A is difficult to predict from the representation. More importantly, removing sensitive information alone does not enlarge the separation between ID and OOD predictive distributions. As a result, OOD samples affected by FS may still be assigned high ID confidence.

These limitations highlight the necessity of a dedicated solution that jointly addresses the direct impact of sensitive attributes and the indirect impact of their induced feature shifts, rather than simply extending existing debiasing methods.

4. PACT: predictive adaptive calibration

To ensure fairness and consequently achieve more reliable OOD detection, we propose a novel approach, called *Predictive Adaptive Calibration* (PACT), comprises two aspects : **Feature Distribution Regularization** (FDR) and **Predictive Distribution Calibration** (PDC).

4.1. FDR: feature distribution regularization

For FDR, by constraining the model to extract highly compact feature representations for ID data while maximizing the distance between ID and OOD features, we compel the model to focus exclusively on class-related features while disregarding sensitive attributes and their induced shifted features. Specifically, this is achieved by minimizing:

$$\mathcal{L}_{\mathrm{FDR}} = -\sum_{\mathbf{x}_j^{\mathrm{I}} \in D_{\mathrm{I}}} \log \frac{\psi(\mathbf{x}_j^{\mathrm{I}}, \mathbf{x}^+)}{\psi(\mathbf{x}_j^{\mathrm{I}}, \mathbf{x}^+) + \sum_{\mathbf{x}^- \in D^-} \psi(\mathbf{x}_j^{\mathrm{I}}, \mathbf{x}^-)}, \tag{7}$$

where $\mathbf{x}^+ \in D^+ = \{\mathbf{x}_t^{\mathrm{I}} | y_t = y_j, z_t = z_j, \ \text{for} \ \forall \mathbf{x}_t^{\mathrm{I}} \in D_{\mathrm{I}}\}$ and the $D^- = \{\mathbf{x}_t^{\mathrm{I}} | y_t \neq y_j, z_t = z_j, \ \text{for} \ \forall \mathbf{x}_t^{\mathrm{I}} \in D_{\mathrm{I}}\} \cup D_{\mathrm{O}}$. And $\psi(\cdot,\cdot) = \exp(\mathrm{sim}(\cdot,\cdot)/\tau)$, where $\mathrm{sim}(\cdot,\cdot)$ denotes the cosine similarity, and τ is the temperature parameter.

For each ID anchor $\boldsymbol{x}_j^I$, let $s_j^+ = \mathrm{sim}(\boldsymbol{x}_j^I, \boldsymbol{x}^+)$ denote its similarity to the positive sample and $s_{jr}^- = \mathrm{sim}(\boldsymbol{x}_j^I, \boldsymbol{x}_r^-)$ denote its similarity to the r-th negative sample. The per-anchor FDR loss can be written as

$$\ell_{\mathrm{FDR}}^j = -\log \frac{\exp(s_j^+/\tau)}{\exp(s_j^+/\tau) + \sum_r \exp(s_{jr}^-/\tau)}. \tag{8}$$

Let q_j^+ and q_{jr}^- denote the corresponding softmax probabilities. The gradients with respect to the positive and negative similarities are

$$\frac{\partial \ell_{\mathrm{FDR}}^j}{\partial s_j^+} = -\frac{1 - q_j^+}{\tau} < 0, \qquad \frac{\partial \ell_{\mathrm{FDR}}^j}{\partial s_{jr}^-} = \frac{q_{jr}^-}{\tau} > 0. \tag{9}$$

Therefore, minimizing $\mathcal{L}_{\mathrm{FDR}}$ increases the similarities between same-class samples and decreases their similarities to samples from different ID classes and OOD data. Since same-class samples with different sensitive attributes are included in the positive set, this optimization reduces sensitive-attribute-related feature variation within each ID class while preserving class and ID–OOD discriminability.

For a class-specific sample set $\mathcal{D}_c$, define

$$\eta_c = \max_{\substack{\mathcal{D}_1, \mathcal{D}_2 \subseteq \mathcal{D}_c \\ \mathcal{D}_1 \neq \mathcal{D}_2}} \left[1 - \mathrm{sim}\left(\mathbb{E}_f(\mathcal{D}_1), \mathbb{E}_f(\mathcal{D}_2)\right)\right] \tag{10}$$

as the maximum discrepancy between the mean representations of two subsets from the same class. By contracting same-class representations, $\mathcal{L}_{\mathrm{FDR}}$ encourages a smaller η_c. When $\eta_c \leq \eta$, the condition in Theorem 1 is satisfied, and the corresponding within-class feature variance is bounded by $\eta^2/4$.

The following theorem characterizes how bounded within-class feature discrepancy constrains the feature variance.

Theorem 1. *Given a feature extractor* $\mathbf{f}$, *let* $D = \{\mathbf{x}_t\}_{t=1}^N$ *be a set of data points belonging to the same class. For any two non-empty distinct subsets* $D_1, D_2 \subseteq D$ *satisfying* $(1 - \mathrm{sim}(E_{\mathbf{f}}(D_1), E_{\mathbf{f}}(D_2))) \leq \eta$, *where the* $E_{\mathbf{f}}(D_j) = \frac{1}{|D_j|}\sum_{\mathbf{x} \in D_j} \mathbf{f}(\mathbf{x})$ *is the mean feature of the subset* D_j $(j \in \{1, 2\})$, *we have:*

$$V_{\mathbf{f}}(D) \leq \frac{\eta^2}{4}, \tag{11}$$

where $V_{\mathbf{f}}(D)$ *denotes feature variance over* D.

Proof. Let μ denote the global mean, i.e.,

$$\mu = \frac{1}{N} \sum_{t=1}^{N} \mathbf{f}(\mathbf{x}_t).$$

Consider singleton subsets $D_1 = \{\mathbf{x}_k\}$ and $D_2 = \{\mathbf{x}_l\}$, where $\mathbf{x}_k$, $\mathbf{x}_l$ are arbitrary samples from the same category. According to $(1 - \mathrm{sim}(E_{\mathbf{f}}(D_1), E_{\mathbf{f}}(D_2))) \leq \eta$, we have:

$$\xi(\mathbf{x}_k, \mathbf{x}_l) = (1 - \mathrm{sim}(E_{\mathbf{f}}(D_1), E_{\mathbf{f}}(D_2))) \leq \eta,$$

where $\xi(\cdot,\cdot)$ is the cosine distance. Thus, the above inequality demonstrates that the feature distance between any two samples does not exceed η. This implies all $\{\mathbf{x}_t\}_{t=1}^{N}$ lie within a closed ball.

Furthermore, since the centroid is a convex combination of points, it must reside within this ball. Consequently, the above conclusion extends to any non-empty, non-singleton subset.

Then, for any $\mathbf{x}_k$ and $\mathbf{x}_l$, we have:

$$\begin{aligned}\xi(\mathbf{f}(\mathbf{x}_k) - \mathbf{f}(\mathbf{x}_l)) &\leq \xi(\mathbf{f}(\mathbf{x}_k) - \mu) + \xi(\mathbf{f}(\mathbf{x}_l) - \mu) \\ &\leq 2\max\{\xi(\mathbf{f}(\mathbf{x}_k) - \mu), \xi(\mathbf{f}(\mathbf{x}_l) - \mu)\}.\end{aligned}$$

According to $\xi(\mathbf{f}(\mathbf{x}_k) - \mathbf{f}(\mathbf{x}_l)) \leq \eta$, we have:

$$2\max\{\xi(\mathbf{f}(\mathbf{x}_k) - \mu), \xi(\mathbf{f}(\mathbf{x}_l) - \mu)\} \leq \eta$$

$$\max\{\xi(\mathbf{f}(\mathbf{x}_k) - \mu), \xi(\mathbf{f}(\mathbf{x}_l) - \mu)\} \leq \frac{\eta}{2}.$$

Since $\mathbf{x}_k$ and $\mathbf{x}_l$ are arbitrary, it follows that:

$$\xi(\mathbf{f}(\mathbf{x}_t), \mu) \leq \frac{\eta}{2}, \forall \mathbf{x}_t \in D.$$

Then, we have:

$$V_{\mathbf{f}}(D) = \frac{1}{N}\sum_{t=1}^{N} \xi(\mathbf{f}(\mathbf{x}_t), \mu)^2 \leq \frac{1}{N}\sum_{t=1}^{N} (\frac{\eta}{2})^2 \leq \frac{\eta^2}{4}.$$

Thus, the proof is completed. □

Remark 3. Minimizing $\mathcal{L}_{\mathrm{FDR}}$ reduces the discrepancy between same-class representations, including those associated with different sensitive attributes, thereby encouraging a smaller η_c. According to Theorem 1, a smaller upper bound on the within-class discrepancy leads to a smaller upper bound on the feature variance. Consequently, FDR suppresses sensitive-attribute-related feature variation and promotes compact representations within each ID class.

4.2. PDC: predictive distribution calibration

PDC is designed to mitigate the indirect influence of feature shifts on OOD decisions. Feature shifts may move the prediction distributions of OOD samples toward the ID predictive region, causing them to receive high ID confidence. Ideally, prediction consistency could be directly enforced between OOD samples with and without FS. However, the FS labels of OOD data are unavailable in practical settings, making it infeasible to construct such paired consistency constraints.

PDC therefore enlarges the predictive separation between correctly classified ID samples and OOD samples. A larger ID–OOD separation provides a predictive margin against the variation introduced by FS and reduces the likelihood that feature-shifted OOD samples produce ID-like predictions. Specifically, this objective is implemented by minimizing:

$$\mathcal{L}_{\mathrm{PDC}} = -\sum_{\mathbf{x}_j^{\mathrm{I}} \in D_{\mathrm{I}}} \sum_{\mathbf{x}_k^{\mathrm{O}} \in D_{\mathrm{O}}} \mathbf{1}\{\hat{y}_j = y_j^{\mathrm{I}}\} \cdot \mathtt{WD}(\mathbf{h}(\mathbf{f}(\mathbf{x}_j^{\mathrm{I}})), \mathbf{h}(\mathbf{f}(\mathbf{x}_k^{\mathrm{O}})) \tag{12}$$

where $\hat{y}_j = \arg\max_y \mathbf{h}^y(\mathbf{f}(\mathbf{x}_j^{\mathrm{I}}))$, and $\mathtt{WD}(\cdot,\cdot)$ denotes the Wasserstein distance.

The following theorem characterizes the variation in prediction distributions induced by a bounded feature shift.

Theorem 2. *Given the feature extractor* $\mathbf{f}$ *and prediction function* $\mathbf{h}$, *and let* $\mathbf{x}' = \mathbf{x} + \delta$ *denotes the feature-shifted input. We assume Wasserstein distance satisfies* $\mathtt{WD}(\mathbf{h}\circ\mathbf{f}(\mathbf{x}), \mathbf{h}\circ\mathbf{f}(\mathbf{x}')) \leq L\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}')\|$, *where* L *is the Lipschitz constant of* $\mathbf{h}$. *When fine-tuning* $\mathbf{f}$ *and* $\mathbf{h}$ *by optimizing the loss* $\mathcal{L}_{PDC}$, *we have: The prediction distribution remains stable under all perturbations* δ *with* $\|\delta\| \leq \epsilon$ *such that:*

$$\|\mathbf{h}\circ\mathbf{f}(\mathbf{x}) - \mathbf{h}\circ\mathbf{f}(\mathbf{x}')\| \leq \epsilon L L_f, \tag{13}$$

where L_f *is the Lipschitz constant of* $\mathbf{f}$.

Proof. For a finite-dimensional neural network with weight matrix W, all singular values of W are finite, and thus its spectral norm is bounded. Since the Lipschitz constant L of the network is given by the spectral norm (Miyato et al., 2018), it follows that L is also bounded. Therefore, the Lipschitz constant is explicitly constrained. According to $\mathtt{WD}(\mathbf{h}\circ\mathbf{f}(\mathbf{x}), \mathbf{h}\circ\mathbf{f}(\mathbf{x}')) \leq L\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}')\|$, as $\mathtt{WD}(\mathbf{h}\circ\mathbf{f}(\mathbf{x}), \mathbf{h}\circ\mathbf{f}(\mathbf{x}'))$ increases, it asymptotically approaches its upper bound $L\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}')\|$.

Given that $\mathbf{x}' = \mathbf{x} + \delta$, where $\|\delta\| \leq \epsilon$, by applying the Lipschitz condition, we obtain:

$$\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}')\| \leq L_f\|\mathbf{x} - \mathbf{x}'\| = L_f\|\delta\| \leq L_f\epsilon.$$

Then, we have:

$$\mathtt{WD}(\mathbf{h}\circ\mathbf{f}(\mathbf{x}), \mathbf{h}\circ\mathbf{f}(\mathbf{x}')) \leq L\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{x}')\| \leq \epsilon L L_f.$$

Thus, the proof is completed. □

Let $p(x) = h(f(x))$ denote the prediction distribution, and define

$$M(\boldsymbol{x}^I, \boldsymbol{x}^O) = \mathrm{WD}\left(p(\boldsymbol{x}^I), p(\boldsymbol{x}^O)\right) \tag{14}$$

as the predictive separation between a correctly classified ID sample $\boldsymbol{x}^I$ and an OOD sample $\boldsymbol{x}^O$. For a feature-shifted OOD sample $\boldsymbol{x}^{O\prime} = \boldsymbol{x}^O + \boldsymbol{\delta}$, where $\|\delta\| \leq \epsilon$, the triangle inequality of the Wasserstein distance gives

$$\mathrm{WD}\left(p(\boldsymbol{x}^I), p(\boldsymbol{x}^{O\prime})\right) \geq M(\boldsymbol{x}^I, \boldsymbol{x}^O) - \mathrm{WD}\left(p(\boldsymbol{x}^O), p(\boldsymbol{x}^{O\prime})\right).$$

According to Theorem 2,

$$\mathrm{WD}\left(p(\boldsymbol{x}^O), p(\boldsymbol{x}^{O\prime})\right) \leq \epsilon L L_f. \tag{15}$$

Therefore,

$$\mathrm{WD}\left(p(\boldsymbol{x}^I), p(\boldsymbol{x}^{O\prime})\right) \geq M(\boldsymbol{x}^I, \boldsymbol{x}^O) - \epsilon L L_f. \tag{16}$$

Minimizing $\mathcal{L}_{\mathrm{PDC}}$ enlarges $M(\boldsymbol{x}^I, \boldsymbol{x}^O)$, thereby increasing the lower bound on the ID–OOD predictive separation that remains after FS. Under a fixed perturbation budget and fixed Lipschitz constants, the relative degradation of the original predictive margin is bounded by $\frac{\epsilon L L_f}{M(\boldsymbol{x}^I, \boldsymbol{x}^O)}$. Increasing the original predictive margin through $\mathcal{L}_{\mathrm{PDC}}$ decreases this relative upper bound. Equivalently, the ID–OOD predictive separation remains positive when $\epsilon < \frac{M(\boldsymbol{x}^I, \boldsymbol{x}^O)}{L L_f}$, indicating that a larger predictive margin permits a larger feature shift without eliminating the separation between ID and OOD samples.

Remark 4. $\mathcal{L}_{\mathrm{PDC}}$ enlarges the original predictive margin $M(\boldsymbol{x}^I, \boldsymbol{x}^O)$ between correctly classified ID samples and OOD samples, while Theorem 2 bounds the prediction variation induced by FS. Their combination yields a residual predictive separation lower bound of $M(\boldsymbol{x}^I, \boldsymbol{x}^O) - \epsilon L L_f$. A larger original margin therefore preserves a larger ID–OOD separation after FS, decreases the relative perturbation bound $\epsilon L L_f / M(\boldsymbol{x}^I, \boldsymbol{x}^O)$, and increases the tolerable feature-shift radius $M(\boldsymbol{x}^I, \boldsymbol{x}^O)/(L L_f)$. Consequently, feature-shifted OOD samples are less likely to produce predictions similar to those of ID samples.

4.3. Overall learning objective

We optimize the classification task using cross-entropy loss ℓ, i.e., minimizing

$$\mathcal{L}_{\mathrm{ce}} = \frac{1}{n}\sum_{\mathbf{x}_j^{\mathrm{I}}} \ell(\mathbf{h}(\mathbf{f}(\mathbf{x}_j^{\mathrm{I}}), y_j^{\mathrm{I}})). \tag{17}$$

Following Outlier Exposure (Hendrycks et al., 2019), we enhance the ability of the model to detect OOD data by minimizing the cross-entropy between model predictions on OOD data and a uniform distribution, i.e.,

$$\mathcal{L}_{\mathrm{oe}} = -\frac{1}{2m}\sum_{\mathbf{x}_k^{\mathrm{O}} \in D_{\mathrm{O}}} \frac{1}{c}\log \mathbf{h}(\mathbf{f}(\mathbf{x}_k^{\mathrm{O}})). \tag{18}$$

Consequently, the overall learning objective is :

$$\mathcal{L} = \mathcal{L}_{\mathrm{ce}} + \mathcal{L}_{\mathrm{oe}} + \alpha\mathcal{L}_{\mathrm{FDR}} + \beta\mathcal{L}_{\mathrm{PDC}}. \tag{19}$$

Algorithm 1 Predictive adaptive calibration (PACT).

Input: The ID training dataset D_{I}; the training OOD dataset D_{O}; the pre-trained model with extractor $\mathbf{f}$ and prediction function $\mathbf{h}$.
Parameter: The trade-off parameters α and β; the temperature parameter τ.
Output: The well-trained model.

```
# Training:
for epoch ← 1 to epochs do
    Fetch a mini-batch (B^I_a, B^I_a', B^O);
    Compute the feature distribution regularization term L_FDR in Eq. (7);
    Compute the predictive distribution calibration term L_PDC in Eq. (12);
    Compute L_ce and L_oe in Eqs. (17) and (18);
    Train the model by minimizing L in Eq. (19));
end for
return The well-trained model.
```

The overall algorithm of our proposed PACT is presented in Algorithm 1. During each training epoch, a minibatch is randomly sampled from the training data, and the model is optimized through a composite objective function that integrates several key components. First, the feature distribution regularization term, which mitigates sensitive attribute biases, is computed using Eq. (7), while the predictive distribution calibration term, which addresses feature shift, is computed using Eq. (12). Furthermore, Eq. (17) optimizes multi-class classification on ID data via cross-entropy loss, while Eq. (18) performs binary classification to separate ID and OOD samples. These components are aggregated into a unified training objective in Eq. (19), enabling end-to-end optimization.

$\mathcal{L}_{\mathrm{FDR}}$ and $\mathcal{L}_{\mathrm{PDC}}$ address the two sources of fairness confusion in a complementary manner. $\mathcal{L}_{\mathrm{FDR}}$ reduces the discrepancy between same-class representations across sensitive groups, thereby encouraging the compactness condition in Theorem 1 and tightening the corresponding feature-variance bound. $\mathcal{L}_{\mathrm{PDC}}$ enlarges the predictive margin between ID and OOD samples, while Theorem 2 bounds the degradation of this margin under FS. Increasing the original predictive margin preserves a larger residual separation and increases the magnitude of FS that can be tolerated without eliminating ID–OOD discriminability. Together, FDR and PDC mitigate the direct influence of sensitive attributes and the indirect influence of their induced feature shifts on OOD decisions.

5. Experiments

In this section, we conduct extensive experiments to evaluate our proposed metric, Fair-OOD, and method, PACT, by answering the following questions:

- **Q1:** Does Fair-OOD effectively identify FC in reality?
- **Q2:** Does PACT enhance the OOD detection performance of the model?
- **Q3:** Does PACT mitigate bias and FC while preserving high classification accuracy?
- **Q4:** Does debiasing with Fair-OOD conflict with other fairness metrics?
- **Q5:** Do the two key components of PACT perform as anticipated in their intended roles?
- **Q6:** Can our PACT benefit other OOD scoring?

5.1. Experiment setup

Dataset. We conduct experiments by employ the datasets: `CIFAR-10-C` (Hendrycks & Dietterich, 2018), `ImageNet-100-C` (Hendrycks & Dietterich, 2018), and `BAR` (Nam et al., 2020). In the `BAR`, each class has its own sensitive attributes. `CIFAR-10-C` assigns the same sensitive attribute to all classes. `ImageNet-100-C` represents a more complex and large-scale setting, comprising numerous classes with high-resolution images. For datasets with predefined training and test splits, we follow the original splits. For datasets without predefined splits, we randomly use 70% of the data for training and the remaining 30% for testing. Specifically, for `BAR`, we use 70% of both the ID and OOD data for training and the remaining 30% for testing. For `CIFAR-100-C`, we follow the original training and test split, while for `ImageNet-100-C`, we use the original training split for training and the validation split for testing. The training and test sets are completely disjoint, with no overlapping samples. Please refer to Appendix B.3 for more details.

Baseline. We compare our proposed method, PACT, with three categories of baseline methods: (1) post-hoc methods in OOD detection, including `MLP` (Hendrycks et al., 2022), `MSP` (Hendrycks & Gimpel, 2017), `T2FNorm` (Regmi et al., 2024a), `Energy` (Liu et al., 2020), `ITP` (Xu & Yang, 2025), and `SCALE` (Xu et al., 2024); (2) fine-tuning methods in OOD detection with additional auxiliary OOD data in OOD detection, including `OE` (Hendrycks et al., 2019), `Energy-OE` (Liu et al., 2020), `DAL` (Wang et al., 2023a), `DiverseMix` (Yao et al., 2024), and `MVOL` (Lei et al., 2025); and (3) fairness-aware representation learning methods, including `DFA` (Lee et al., 2021), `SelecMix` (Hwang et al., 2022), `BCSI` (Jung et al., 2024), `DPR` (Han et al., 2024), and `ALFA` (Jung et al., 2025).

Metric. We evaluate the OOD detection performance and fairness using the following metrics:

- For OOD detection: the false positive rate at which OOD samples are declared as ID when 95% of ID samples are declared as ID (FPR95); and the area under the receiver operating characteristic curve (AUROC);
- For ID classification: the classification accuracy on ID data with sensitive attribute $A = a$ (ACC-U) and on ID data with $A = a'$ (ACC-B);
- For Fairness: Demographic Parity (DP) computed by $\mathrm{DP} = |P(\hat{Z} = o|A = a) - P(\hat{Z} = o|A = a')|$; Equal Opportunity (EO) computed by $\mathrm{EO} = |P(\hat{Z} = o|A = a, Z = o) - P(\hat{Z} = o|A = a', Z = o)|$; and Equalized Odds (EOD) computed by $\mathrm{EOD} = \frac{1}{2}|P(\hat{Z} = o|A = a, Z = o) - P(\hat{Z} = o|A = a', Z = o)| + |P(\hat{Z} = o|A = a, Z = i) - P(\hat{Z} = o|A = a', Z = i)|$; and our Fair-OOD computed by

$$
\begin{aligned}
\text{Fair-OOD} = \frac{1}{4} \sum_{z \in \{i,o\}} \Big\{ & |P(\hat{Z} = \hat{z}|A = a, S = s'_a, Z = z) \\
& - P(\hat{Z} = \hat{z}|A = a', S = s'_{a'}, Z = z)| \\
& + |P(\hat{Z} = \hat{z}|A = a', S = s_{a'}, Z = z) \\
& - P(\hat{Z} = \hat{z}|A = a', S = s'_{a'}, Z = z)| \Big\}.
\end{aligned}
$$

where $z = \hat{z}$, and the values of S are determined by the same scheme in our case study.

PACT fine-tuning setup. We employ ResNet-18 (He et al., 2016) as the backbone model and fine-tune it using stochastic gradient descent with Nesterov momentum (Duchi et al., 2011). The weight decay coefficient, momentum, and initial learning rate are set to 0.0005, 0.09, and 0.0001, respectively. We fine-tune the model for 3 epochs on `BAR` and `CIFAR-10-C` and for 10 epochs on `ImageNet-100-C`, owing to the larger scale and greater complexity of `ImageNet-100-C`. For all datasets, the batch sizes of both ID and auxiliary OOD data are set to 128. The contrastive temperature is set to $\tau = 0.1$, while the loss weights of $\mathcal{L}_{\mathrm{FDR}}$ and $\mathcal{L}_{\mathrm{PDC}}$ are set to $\alpha = 2$ and $\beta = 2$, respectively. The MLP score is used as the OOD score during evaluation. The complete pre-training and fine-tuning configurations are provided in Appendix B.2.

5.2. Main result

We primarily highlight the following observations:

(a) **Fair-OOD captures the FC in OOD detection (Q1).** As shown in Table 1, prior to applying PACT, Fair-OOD remain significantly higher

Table 1
Comparison between our proposed PACT and advanced methods across different datasets. For each dataset, we perform a categorical division into ID and OOD partitions. ↑ (or ↓) indicates larger or smaller values are preferred. Results are reported as percentage values, averaged over 10 runs with $_{\pm q}$ denoting the standard error. Bold font indicates the best results in a column.

Method	Classification		Fairness				OOD Detection	
	ACC-U ↑	ACC-B ↑	DP ↓	EO ↓	EOD ↓	Fair-OOD ↓	FPR95 ↓	AUROC ↑
BAR								
MSP	85.42	50.85	39.44	73.72	67.03	27.10	84.53	63.07
MLP	85.42	50.85	40.65	59.33	37.01	17.97	69.54	60.77
T2FNorm	85.67	54.09	40.39	70.92	70.50	17.81	72.07	63.14
Energy	85.42	50.85	40.95	79.32	70.21	18.72	70.94	65.17
ITP	85.42	50.85	38.91	72.73	66.10	17.05	67.15	64.21
SCALE	85.42	50.85	39.35	70.76	65.17	16.97	65.22	65.22
DFA	83.01	56.10	35.19	55.10	67.59	15.80	87.44	63.83
SelecMix	85.67	57.96	36.09	31.20	65.60	15.90	82.13	59.70
BCSI	83.29	57.61	33.72	33.72	60.59	15.07	79.01	61.29
DPR	84.41	63.41	34.87	27.52	52.33	12.32	77.01	62.51
ALFA	84.30	66.12	34.63	24.60	47.64	10.39	75.29	63.20
PACT (Ours)	**85.94**$_{\pm 0.2}$	**77.48**$_{\pm 0.6}$	**33.47**$_{\pm 1.7}$	**16.67**$_{\pm 0.5}$	**25.16**$_{\pm 1.2}$	**4.17**$_{\pm 0.0}$	**60.99**$_{\pm 0.7}$	**65.28**$_{\pm 0.5}$
CIFAR-10-C								
MSP	81.29	69.11	9.34	12.90	7.74	6.55	88.02	60.57
MLP	81.29	69.11	7.86	5.06	6.69	5.63	84.55	63.04
T2FNorm	81.31	69.09	5.28	4.19	4.06	9.88	82.76	64.29
Energy	81.29	69.11	9.42	5.61	7.83	5.27	87.44	61.41
ITP	81.29	69.11	7.12	4.14	6.92	5.53	76.21	69.60
SCALE	81.29	69.11	6.25	4.14	5.39	4.43	73.19	72.61
DFA	81.22	68.70	4.73	2.95	3.94	5.00	87.14	62.49
SelecMix	81.51	68.76	5.71	2.67	5.17	5.92	85.17	60.07
BCSI	81.09	70.18	3.97	2.62	5.65	6.77	81.19	65.97
DPR	81.01	72.61	3.12	2.62	4.10	4.41	80.76	65.70
ALFA	80.10	73.42	2.59	2.77	3.71	3.95	78.31	66.92
PACT (Ours)	**81.55**$_{\pm 0.2}$	**78.27**$_{\pm 0.3}$	**0.50**$_{\pm 0.0}$	**2.48**$_{\pm 0.1}$	**1.30**$_{\pm 0.2}$	**1.59**$_{\pm 0.1}$	**38.79**$_{\pm 0.3}$	**89.50**$_{\pm 0.2}$

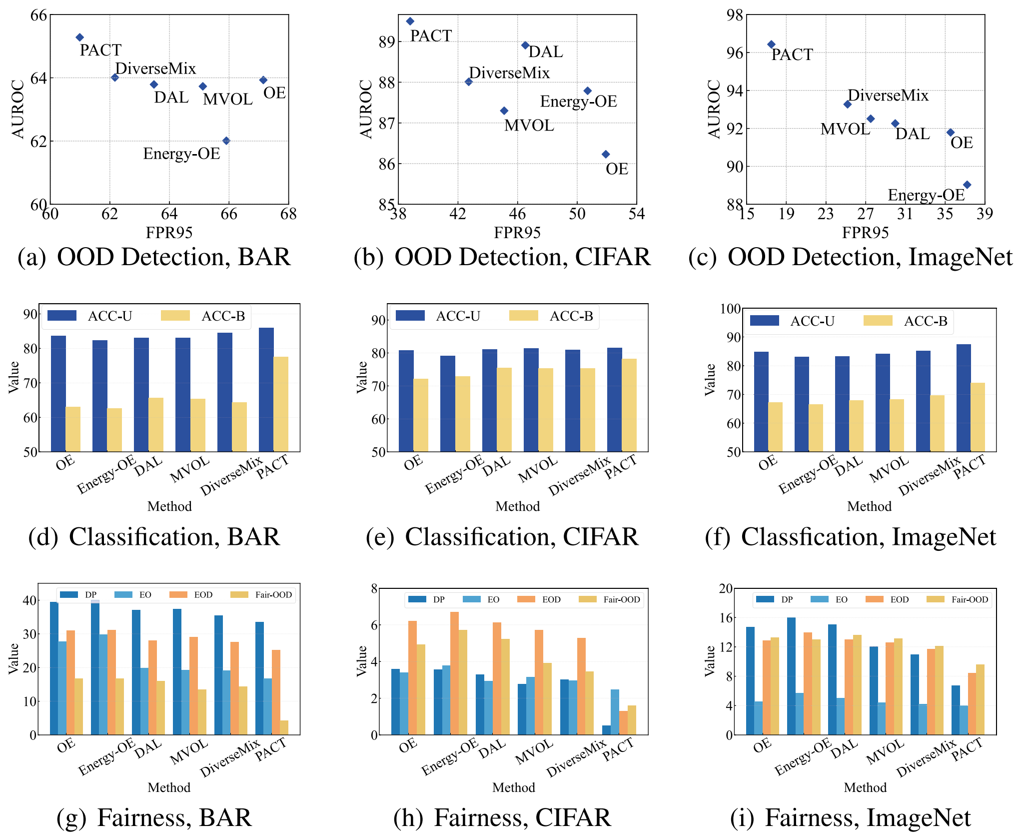
Table 2
Comparison between our proposed PACT and advanced methods on challenge ImageNet-100-C. For each dataset, we perform a categorical division into ID and OOD partitions. ↑ (or ↓) indicates larger or smaller values are preferred. Results are reported as percentage values, averaged over 10 runs with $_{\pm q}$ denoting the standard error. Bold font indicates the best results in a column.

Method	Classification		Fairness				OOD Detection	
	ACC-U ↑	ACC-B ↑	DP ↓	EO ↓	EOD ↓	Fair-OOD ↓	FPR95 ↓	AUROC ↑
MSP	81.89	62.96	24.51	5.27	13.63	15.28	61.16	76.80
MLP	81.89	62.96	25.72	4.02	19.84	15.50	56.81	78.71
T2FNorm	81.70	61.59	25.10	5.92	18.50	17.75	66.03	75.25
Energy	81.89	62.96	25.99	**3.88**	19.99	17.50	56.63	78.33
ITP	81.89	62.96	25.26	4.76	18.11	14.63	49.93	82.62
SCALE	81.89	62.96	24.43	4.69	17.13	14.43	46.41	84.97
DFA	81.22	63.71	7.84	6.97	13.39	15.24	87.43	61.39
SelecMix	83.31	65.92	7.03	5.25	12.12	15.90	85.77	62.70
BCSI	83.93	65.70	6.92	5.09	12.60	13.71	81.70	65.17
DPR	84.11	67.42	6.26	4.87	11.83	12.64	78.27	66.87
ALFA	84.63	86.94	5.96	4.69	10.27	11.85	76.97	68.27
PACT (Ours)	**87.41**$_{\pm 0.5}$	**74.03**$_{\pm 0.3}$	**6.69**$_{\pm 0.2}$	3.96$_{\pm 0.2}$	**8.43**$_{\pm 0.7}$	**9.56**$_{\pm 0.2}$	**17.50**$_{\pm 0.3}$	**96.43**$_{\pm 0.2}$

across all datasets. For instance, the `MSP` yields a Fair-OOD score of 27.10% on the `BAR` dataset. This observation aligns with our case study findings that OOD detection is not only influenced by sensitive attributes but also exhibits FC.

(b) **PACT significantly improves OOD detection performance (Q2).** Across all datasets, our proposed PACT framework demonstrates significant improvements in OOD detection performance. Most notably, on the large-scale, multi-class ImageNet-100-C benchmark, as shown in Table 2, PACT achieves a substantial 11.46% absolute improvement in AUROC over the strongest baseline. In contrast, existing debiasing methods show negligible performance gains in OOD detection, exhibiting consistently weak detection capabilities. These results validate reliable performance of our PACT.

(c) **PACT not only mitigates bias and resolves fairness confusion, but also improves classification accuracy (Q3).** As evidenced in Table 1, conventional debiasing methods demonstrate competent performance on standard fairness metrics but consistently fail to optimize Fair-OOD. In contrast, our PACT simultaneously optimizes both conventional fairness metrics and Fair-OOD, effectively mitigating FC. This also demonstrates that **debiasing with Fair-OOD is not in conflict with conventional fairness metrics (Q4).** Furthermore, while existing debiasing methods and prior OOD detection approaches exhibit significant



(a) OOD Detection, BAR (b) OOD Detection, CIFAR (c) OOD Detection, ImageNet

(d) Classification, BAR (e) Classification, CIFAR (f) Classfication, ImageNet

(g) Fairness, BAR (h) Fairness, CIFAR (i) Fairness, ImageNet

Fig. 4. Comparison between our proposed PACT and advanced methods using additional auxiliary OOD data across three datasets. .

Table 3
Ablation study of the key components of PACT on BAR, CIFAR-10-C, and ImageNet-100-C, where Average ACC is computed as the arithmetic mean of ACC-U and ACC-B.

Index	Component				BAR			CIFAR-10-C			ImageNet-100-C		
	$\mathcal{L}_{\text{FDR}}$	$\mathcal{L}_{\text{SupCon}}$	$\mathcal{L}_{\text{PDC}}$	$\mathcal{L}_{\text{Euc}}$	Average ACC↑	Fair-OOD↓	FPR95↓	Average ACC↑	Fair-OOD↓	FPR95↓	Average ACC↑	Fair-OOD↓	FPR95↓
1					72.42	25.00	65.98	74.18	9.93	54.85	74.10	12.25	33.94
2		✓			75.27	33.33	64.71	79.11	3.31	54.33	72.33	8.82	30.06
3	✓				79.07	16.67	64.44	79.21	2.82	48.01	78.51	11.27	22.61
4				✓	71.81	29.17	65.31	78.28	5.27	59.53	76.65	12.01	29.61
5			✓		72.68	8.33	61.57	78.16	1.76	47.46	78.55	8.33	20.74
6		✓		✓	74.23	14.29	65.58	78.94	3.06	58.83	77.13	10.78	38.14
7	✓		✓		**81.71**	**4.17**	**60.99**	**79.91**	**1.59**	**38.79**	**80.72**	**9.56**	**17.50**

disparities in classification accuracy across different sensitive attribute values, PACT-optimized models achieve substantially closer ACC-U and ACC-B values.

5.3. In-depth analysis

FDR and PDC jointly mitigate both sensitive-attribute-induced fairness issues and feature-shift-induced fairness confusion (Q5). To determine whether the improvements arise from the specific designs of FDR and PDC rather than generic additional regularization, we introduce two simplified alternatives. $\mathcal{L}_{\text{SupCon}}$ serves as a simpler class-aware contrastive baseline that promotes compact representations within each ID class, while $\mathcal{L}_{\text{Euc}}$ replaces the Wasserstein distance in PDC with the Euclidean distance between prediction vectors. All variants use the same backbone, data splits, base objectives, training protocol, and hyperparameter-selection procedure.

From Table 3, both FDR and PDC consistently improve the base model across `BAR`, `CIFAR-10-C`, and `ImageNet-100-C`, while their combination achieves the best overall performance. Compared with standard supervised contrastive regularization, FDR achieves a better overall trade-off among Average ACC, Fair-OOD, and FPR95, highlighting the effectiveness of its OOD-aware negative construction. PDC also consistently obtains lower Fair-OOD and FPR95 than the Euclidean-distance alternative, supporting the use of Wasserstein distance for predictive distribution calibration. Moreover, full PACT outperforms the combination of the two simplified alternatives across all three datasets, indicating that its improvements cannot be fully reproduced by generic contrastive and prediction-separation regularization and instead benefit from the specific formulations of FDR and PDC.

Similarly, PDC primarily targets feature-shift-induced fairness confusion, reducing Fair-OOD from 25.00% to 8.33% on BAR. In addition, it also improves the average accuracy from 72.42% to 72.68% and reduces FPR95 from 65.98% to 61.57%, suggesting that PDC can further help alleviate fairness issues associated with sensitive attributes. When FDR and PDC are combined, PACT achieves the best overall performance, with the highest average accuracy and the lowest Fair-OOD and FPR95. These results demonstrate that FDR and PDC are complementary: each component optimizes its intended objective while simultaneously help-

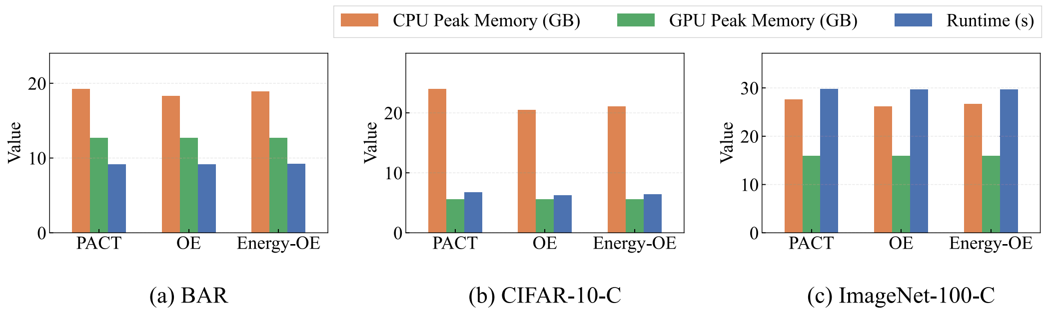


Fig. 5. Computational overhead comparison on BAR, CIFAR-10-C, and ImageNet-100-C. We compare CPU peak memory, GPU peak memory, and runtime across different methods. CPU and GPU memory costs are reported in GB. For ImageNet-100-C, the runtime values are divided by 10 for visualization only, while the original runtime values are used in the quantitative analysis.

ing mitigate the other fairness-related challenge. When combined, they further reinforce each other and enable PACT to achieve the best overall performance.

Employing PACT to optimize models can benefit other OOD scoring methods (Q6). From Fig. 6(a), PACT optimization significantly reduces Fair-OOD metric values across all OOD scoring methods. This observation substantiates both the effectiveness and general applicability of PACT across different OOD detection approaches.

Comparison with methods using additional auxiliary OOD data. To further investigate the performance of our proposed PACT, we conduct additional comparisons with methods that fine-tune models using auxiliary OOD data, including `OE`, `Energy-OE`, `DAL`, `MVOL`, and `DiverseMix`. The experimental results are shown in Fig. 4. On the one hand, our proposed Fair-OOD reveals the fairness confusion issue existing in the compared methods. On the other hand, the model fine-tuned with PACT effectively mitigates the fairness confusion problem while maintaining correct classification on ID data and reliable detection of OOD data. Moreover, PACT achieves the best performance across all datasets, further demonstrating the effectiveness of our proposed method.

Parameter analysis. As shown in Fig. 6(b), PACT consistently outperforms the state-of-the-art (SOTA) method across a broad range of parameter configurations: $\beta \in [0, 6]$, $\tau \in [0.1, 0.5]$, and all values of α. This demonstrates that PACT maintains robust performance without requiring meticulous fine-tuning, indicating its strong generalization capability across diverse hyperparameter settings.

Computational overhead analysis. We further evaluate the computational overhead of different methods on `ImageNet-100-C`, `CIFAR-10-C`, and `BAR` in terms of three metrics: runtime, CPU peak memory, and GPU peak memory.

As shown in Fig. 5, the three methods exhibit comparable computational costs across datasets. On `ImageNet-100-C`, PACT, `OE`, and `Energy` require nearly identical runtimes, around 296–298 seconds, and their GPU memory usages are also almost the same, approximately 15.95 GB. PACT has slightly higher CPU memory usage than `OE` and `Energy`, but the difference remains moderate. On CIFAR-10-C and BAR, the same trend can be observed. GPU memory consumption is nearly unchanged across the three methods, indicating that the compared methods have similar GPU memory footprints. The runtime differences are also small, especially on BAR, where all methods finish within around 9.1 seconds. Although PACT introduces a slightly higher CPU memory increase in some cases, its runtime and GPU memory cost remain comparable to `OE` and `Energy`. These results demonstrate that PACT delivers substantial performance gains while incurring negligible additional computational overhead, with runtime and GPU memory requirements remaining comparable to those of existing methods.

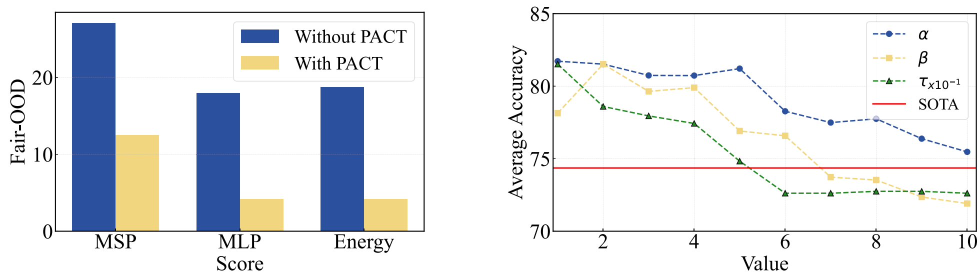
6. Related work

In this section, we review the research in fairness-aware representation learning and OOD detection.

Fairness-aware representation learning. The existing fairness-aware representation learning methods can be primarily categorized into the following approaches: reweighting, data augmentation, feature disentanglement, robust optimization, along with other miscellaneous techniques that do not fall into these main categories. Sample reweighting represents one of the earliest proposed algorithmic approaches in this domain, which mitigates model overfitting to biased attributes in majority groups by assigning higher weights to minority group instances compared to majority group samples (Cheng et al., 2024; Kim et al., 2022; Roh et al., 2023). Data augmentation methods enhance model fairness by improving generalization capabilities across diverse data populations through strategically expanded training set diversity (Kim et al., 2021; Li et al., 2025a; Ramaswamy et al., 2021). Feature disentanglement methods aim to decouple the correlation between target features and bias attributes in the latent representation space, thereby enabling models to learn bias-invariant target features and mitigate interference from spurious biased correlations (Ragonesi et al., 2021; Zhang et al., 2025; Zhu et al., 2021). Robust optimization enhances model fairness by minimizing the worst-case loss over diverse training-set subsets or formally defined perturbation sets surrounding the data distribution (Du & Wu, 2021; Li et al., 2025b; Mandal et al., 2020). In addition, other methods improve model fairness through architectural modifications (Shrestha et al., 2022; Sreelatha et al., 2024) or post-processing techniques applied to the model outputs (Basu et al., 2024). However, existing fairness-aware representation learning methods exhibit critical limitations in OOD detection task, particularly in addressing potential fairness confusion issue.

6.1. Out-of-distribution detection

Out-of-distribution detection Current OOD detection methods can be broadly categorized into three approaches: post-hoc methods, representation-based methods, and the outlier exposure. Post-hoc methods focus on designing various OOD scoring functions to distinguish between ID and OOD data patterns. These approaches leverage different types of information from model outputs. For instance, researchers (Hendrycks & Gimpel, 2017; Liu et al., 2020; Peng et al., 2024) develop OOD scoring functions using the logit outputs of model, while other methods utilize extracted features (Luo et al., 2023; Regmi et al., 2024b; Wang et al., 2022a) or gradient information (Huang et al., 2021; Igoe et al., 2022; Liang et al., 2018) for this purpose. Representation-



(a) Different OOD Scores (b) Hyper-parameter Sensitivity Analysis

Fig. 6. Fig. 6(a) presents Fair-OOD results obtained with different OOD scores before and after training with PACT. Fig. 6(b) illustrates the sensitivity analysis of hyper-parameters α, β, and τ.

based methods posit that discriminative ID representations are pivotal for OOD. They enhance the quality of these representations via contrastive learning (Sehwag et al., 2021; Wang et al., 2022b), data augmentation (Tack et al., 2020), or regularization imposed on embedding features and model outputs (Du et al., 2022; Ming et al., 2023; Wei et al., 2022; Zhang et al., 2024). Outlier exposure augments model training with auxiliary OOD data, thereby equipping the classifier with explicit OOD knowledge and sharpening its discriminative boundary between ID and OOD instances (Hendrycks et al., 2019; Wang et al., 2023a, 2025, 2023b,c). However, current OOD detection methods neglect fairness considerations, potentially degrading both detection performance and model reliability.

7. Conclusion

In this paper, we incorporate fairness considerations into OOD detection and introduce a challenging problem, *Fair OOD Detection*, which involves biases induced by sensitive attributes and *Fairness Confusion* (FC) arising from *Feature Shifts* (FS) triggered by these attributes. To quantify FC, we propose a novel metric, Fair-OOD. Furthermore, we propose *Predictive Adaptive Calibration* (PACT), a method that jointly enhances OOD detection performance, mitigates bias, and alleviates FC. Extensive experiments validate the effectiveness of Fair-OOD in identifying FC and the efficacy of PACT in delivering robust and fairness-aware OOD detection. We hope our work will inspire future research on OOD detection and fairness-aware representation learning.

CRediT authorship contribution statement

Zhaohui Hu: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization; **Haotian Wang:** Formal analysis; **Jialu Zhou:** Validation; **Chuan Li:** Visualization; **Da Huang:** Writing – review & editing; **Long Lan:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Details on comparison between our Fair-OOD and other fairness notion

To formally distinguish between Fair-OOD and DP, we construct a counterexample. To make it more intuitive, we present the unfair decision model in Table B.4, along with $P(S=s)=P(S=s')=0.5$, $P(S=s'|A=a)=0.99$, and $P(S=s'|A=a')=0.01$. Then, we derive the DP as follows.

$$\begin{aligned} &P(\hat{Z}=o|A=a) \\ &=P(\hat{Z}=o|S=s,A=a)P(S=s|A=a) \\ &\quad+P(\hat{Z}=o|S=s',A=a)P(S=s'|A=a) \\ &=0.001+0.495=0.496\approx 0.5. \end{aligned} \tag{A.1}$$

$$\begin{aligned} &P(\hat{Z}=o|A=a') \\ &=P(\hat{Z}=o|S=s,A=a')P(S=s|A=a') \\ &\quad+P(\hat{Z}=o|S=s',A=a')P(S=s'|A=a') \\ &=0.495+0.009=0.504\approx 0.5. \end{aligned} \tag{A.2}$$

Consequently, we obtain that $P(\hat{Z}=o|A=a)\approx 0.5=0.5\approx P(\hat{Z}=o|A=a')$. Therefore, DP cannot detect unfairness in this scenario, where the bias can also be induced by feature shift, but our Fair-OOD can.

Appendix B. Additional experimental details

In this section, we present details for the empirical study and experiments on the main body.

B.1. Additional details of empirical study

Our case study is conducted by employing `ResNet-18` as the model. We train the model to ensure fair classification performance on ID data. The distance metric is the Euclidean distance.

Furthermore, we determine the values of S using the following approach.

$$S(\mathbf{x}_t)=\begin{cases} s', & \text{if}\min_{y\in\mathcal{Y}}\phi(\mathbf{x}_t,\mu^y)<\tau, \\ s, & \text{if,}\min_{y\in\mathcal{Y}}\phi(\mathbf{x}_t,\mu^y)\geq\tau, \end{cases} \tag{B.1}$$

where $\phi(\cdot,\cdot)$ is a distance metric, μ^y represents the feature centroid corresponding to the y-th class, and we compute it by $\mu^y=\frac{1}{n^y}\sum_{i=1}^{n^y}\mathbf{x}_i^y$, where n^y denotes the number of samples in the y-th class. Since feature-shifted samples typically exhibit larger distances from the original class centroids, samples whose minimum distance to the class centroids is no smaller than τ are assigned $S=s$, indicating the presence of FS. The remaining samples are assigned $S=s'$, indicating the absence of FS. The

Table B.4
Unfair desision model.

A	S	$P(\hat{Z}=o\|A,S)$
a	s	0.1
a	s'	0.5
a'	s	0.5
a'	s'	0.9

Table B.5
Sensitive attributes for each class in BAR.

Class	Sensitive Attribute	
	$A=a$	$A=a'$
Climbing	Rock Wall	Other Background
Diving	Underwater	
Fishing	Water Surface	
Pole Vaulting	Sky	
Racing	A Paved Track	
Throwing	Playing Field	

threshold τ is set as the 70th percentile of the ordered set of ID sample-centroid distances $\phi(\mathbf{x},\mu^y)$, namely,

$$\tau = \mathtt{SORT}_{70}\{\phi(\mathbf{x}_i^y,\mu^y);\ \text{for } \mathbf{x}_i^y \in D_\mathrm{I}\}, \tag{B.2}$$

where $\mathtt{SORT}_{70}\{\cdot\}$ sorts a set to its increasing order and return the 70th percentile element from the results, and the 70th percentile is selected as the threshold because feature-shifted samples typically exhibit larger distances to the original class centroids.

B.2. Additional details of experiments

Pre-training Setups. We employ ResNet-18 (He et al., 2016) as the backbone network. We pre-train the model for 50 epochs with cross-entropy loss. The optimizer is SGD with a momentum of 0.9, an initial learning rate of 0.01, and a cosine annealing decay schedule (Loshchilov & Hutter, 2017). Additionally, a weight decay of 0.0005 is applied.
PACT fine-tuning setups. Starting from the pre-trained model, we perform PACT fine-tuning using SGD with Nesterov momentum. The momentum and weight decay coefficient are set to 0.09 and 0.0005, respectively, for all datasets. The initial learning rate is set to 0.0001 for BAR and CIFAR-10-C and to 0.001 for ImageNet-100-C. We retain the default hyperparameter settings for all baseline methods.

B.3. Datasets

BAR (Nam et al., 2020) is a real-world action recognition dataset. We summarize the sensitive attributes for each class in Table B.5. We designate data belonging to four classes as ID data and utilize data from the remaining two classes as OOD data.

CIFAR-10-C (Hendrycks & Dietterich, 2018) is derived from CIFAR-10 (Krizhevsky & Hinton, 2009) by applying various corruptions, including gaussian noise, defocus blur, glass blur, impulse noise, shot noise, snow, and zoom blur. We use CIFAR-10 as the data with $A=a$, and CIFAR-10-C as the data with $A=a'$. We designate data belonging to seven classes as ID data and utilize data from the remaining three classes as OOD data.

ImageNet-100-C (Hendrycks & Dietterich, 2018) is derived from ImageNet-100 (Deng et al., 2009), which is the subset of ImageNet-1k, containing 100 distinct classes, and consists of multiple types of algorithmically generated corruptions from noise, blur, weather, and digital categories. We use ImageNet-100 as the data with $A=a$, and ImageNet-100-C as the data with $A=a'$. We designate data belonging to 70 classes as ID data and utilize data from the remaining 30 classes as OOD data.

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